

Control work 3, and work on the mistakes

Pyramids, cones and spheres in one paper, then the quarter drawn as a single map.

LESSON 51–52

2 × 40 MIN

GEOMETRY 11, NAZORAT ISHI 3

IGCSE E13 REVIEW

LESSON CLOCK

3

42'

12'

20'

10'

3

Instructions	3 min
The paper	42 min
Answers	12 min
Rewrite	20 min
The map	10 min
Targets	3 min

LEARNING OBJECTIVES

- Apply the formulas of Quarter III in one assessment.
- Choose correctly between slant height, height and lateral edge.
- Classify each lost mark and rewrite the solution in full.
- Draw the quarter as one map, from the pyramid to the sphere.

TEXTBOOK

National	pp. 249–252
Cambridge	Core & Extended, pp. 375–378
Workbook	Control paper G3

01

Explanation

The paper — 35 marks, 42 minutes

Q	TASK	MARKS	FROM
1	Regular square pyramid, base edge 10, height 12: apothem, lateral area, total area	6	L33–34
2	The same pyramid: its volume, and the angle its lateral edge makes with the base	6	L33–34, L43–44
3	A cone of radius 5 and slant 13: height, curved area, volume, and the sector angle of its net	7	L36–38
4	A frustum of a cone, radii 6 and 3, height 4: volume and curved area	6	L39
5	A ball of surface area 144π : its radius and volume; then the volume of a cap of height 2	5	L41–42, L45–47
6	Two similar cones have volumes 27 and 125: the ratios of heights and of surface areas	5	L40

WHERE THE MARKS ARE ACTUALLY LOST

Q1 and Q3 each carry a mark for using the **slant height** in an area and the **vertical height** in a volume. Q4 carries one for not treating the frustum as a prism. Q5 carries one for reading “surface area” rather than “volume”. Four of the thirty-five marks are for reading.

Mark by mark

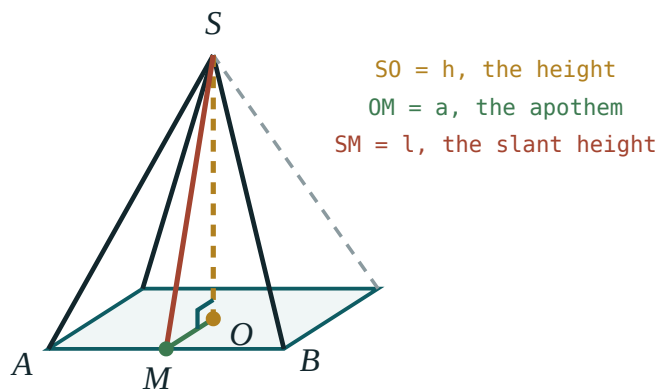
SLIP	WHAT IT LOOKS LIKE	THE FIX
slant for height	$V = \frac{1}{3}Bl$	volume always uses h , the perpendicular
height for slant	$S = \frac{1}{2}Ph$	area always uses m , the apothem
frustum by subtraction only	finding the whole cone first, then failing	use $\frac{\pi h}{3}(R^2 + Rr + r^2)$ directly
the missing $\frac{1}{3}$	$V = Bh$	pyramids and cones are a third of the prism
sphere formulas swapped	$S = \frac{4}{3}\pi R^3$	area has R^2 , volume has R^3
ratios not cubed	$V_1 : V_2 = k$	k, k^2, k^3

For every mark lost, write the slip's name in the margin and then rewrite the whole question — not the line that went wrong. A rewritten line is forgotten by Friday; a rewritten solution is not.

The quarter as one map

Six boxes and the links as sentences:

- **prism** → **pyramid** — “a third of it, on the same base and height”
- **cylinder** → **cone** — “the same third, with $B = \pi r^2$ ”
- **cone** → **frustum** — “cut off the top, or use $R^2 + Rr + r^2$ ”
- **net** → **curved area** — “the cone unrolls into a sector, $S = \pi r l$ ”
- **sphere** → **its cylinder** — “2 : 3 for both volume and area”
- **scale k** → **k, k^2, k^3** — “length, area, volume”



$$l^2 = h^2 + a^2 \quad S_{\text{lat}} = \frac{1}{2}Pl$$

The three lengths of the first lesson — still the whole of Q1 and Q2.

SEVEN FORMULAS, ONE PAGE

$$V_{\text{pyr}} = \frac{1}{3}Bh \quad S_{\text{lat}} = \frac{1}{2}Pl \quad V_{\text{cone}} = \frac{1}{3}\pi r^2 h \quad S_{\text{cone}} = \pi rl$$

$$V_{\text{frus}} = \frac{\pi h}{3}(R^2 + Rr + r^2) \quad V_{\text{sph}} = \frac{4}{3}\pi R^3 \quad S_{\text{sph}} = 4\pi R^2$$

If these seven sit on one page in your own handwriting, Quarter IV needs nothing new.

Looking forward

Quarter IV adds no new solid. It is application: containers, costs, optimisation and the entrance paper, where a question rarely says which formula it wants. The reading skill of Lesson 48–50 becomes the whole of the work.

ONE HABIT TO CARRY FORWARD

Before any arithmetic, write down the object, its decomposition and the letter of every length you know. Almost every mark lost this quarter was lost before the first calculation.

02 Worked examples

EXAMPLE 1 Model answer, Q1: a regular square pyramid, base edge 10, height 12.

① Apothem of the base 5; so $m = \sqrt{144 + 25} = 13$.

The first right triangle.

② $S_{lat} = \frac{1}{2}(40)(13) = 260$.

Perimeter times apothem, halved.

③ $B = 100$.

④ $S_{total} = 360$.

ANSWER

$$m = 13, S_{lat} = 260, S_{total} = 360$$

EXAMPLE 2 Model answer, Q3: a cone of radius 5 and slant 13.

$$① \quad h = \sqrt{169 - 25} = 12.$$

$$② \quad S_{\text{curved}} = \pi(5)(13) = 65\pi.$$

Slant, not height.

$$③ \quad V = \frac{1}{3}\pi(25)(12) = 100\pi.$$

Height, not slant.

$$④ \quad \text{Sector angle } \theta = 360^\circ \times \frac{5}{13} \approx 138.5^\circ.$$

$$\theta = 360^\circ \frac{r}{l}.$$

ANSWER

$$h = 12, 65\pi, 100\pi, \approx 138.5^\circ$$

EXAMPLE 3 Model answer, Q4: a frustum, radii 6 and 3, height 4.

$$① \quad V = \frac{\pi(4)}{3}(36 + 18 + 9) = 84\pi.$$

Use the formula directly.

$$② \quad \text{Slant } l = \sqrt{16 + 9} = 5.$$

From the difference of radii.

$$③ \quad S_{\text{curved}} = \pi(6 + 3)(5) = 45\pi.$$

$$④ \quad V \approx 264, S \approx 141.$$

ANSWER

$$84\pi \text{ and } 45\pi$$

EXAMPLE 4 Model answer, Q6: two similar cones of volumes 27 and 125.

$$① \quad k^3 = \frac{125}{27}$$

$$② \quad k = \frac{5}{3}$$

Heights are in 3 : 5.

$$③ \quad k^2 = \frac{25}{9}$$

④ Surface areas are in 9 : 25.

ANSWER

3 : 5 and 9 : 25

03 Interactive model

Work Q3 on the board twice — once correctly, once with slant and height swapped — and let the class find the difference.

The quarter in ten questions

Volume of a pyramid:

Bh

$\frac{1}{3} Bh$

$\frac{1}{2} Bh$

$3Bh$

Lateral area of a regular pyramid:

$\frac{1}{2} Ph$

$\frac{1}{2} Pm$

Pm

Bm

Curved area of a cone:

πr^2

$\pi r l$

$\pi r h$

$2\pi r l$

Volume of a cone:

$$\pi r^2 h$$

$$\frac{1}{3} \pi r^2 h$$

$$\frac{1}{3} \pi r^2 l$$

$$\pi r l$$

The sector angle of a cone's net:

$$360^\circ \frac{r}{l}$$

$$360^\circ \frac{l}{r}$$

$$180^\circ \frac{r}{l}$$

$$90^\circ$$

Volume of a frustum of a cone:

$$\frac{\pi h}{3} (R^2 + r^2)$$

$$\frac{\pi h}{3} (R^2 + Rr + r^2)$$

$$\pi h (R^2 - r^2)$$

$$\frac{\pi h}{2} (R + r)$$

Volume of a sphere:

$$4\pi R^2$$

$$\frac{4}{3}\pi R^3$$

$$\frac{2}{3}\pi R^3$$

$$\pi R^3$$

Surface area of a sphere:

$$4\pi R^2$$

$$\frac{4}{3}\pi R^3$$

$$2\pi R^2$$

$$\pi R^2$$

A plane section of a sphere is:

an ellipse

a circle

a parabola

a point

Scale factor k multiplies volume by:

k

k^2

k^3

$3k$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Slant height	Apofema	Апофема
Lateral edge	Yon qirra	Боковое ребро
Frustum	Kesik jism	Усечённое тело
Sphere	Sfera	Сфера
Similar solids	O'xshash jismlar	Подобные тела
Concept map	Tushunchalar xaritasi	Карта понятий
Target	Maqsad	Цель

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 needs the apothem because:

it is easier

lateral area uses the slant height

volume uses it

no reason

Q3's volume uses:

the slant l

the height h

the radius only

the sector angle

The frustum formula has:

two terms

three terms

one term

four terms

Volumes $27 : 125$ give heights in:

$27 : 125$

$3 : 5$

$9 : 25$

$1 : 5$

Quarter IV begins with:

new solids

applying these formulas in context

vectors

probability

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Square pyramid, base 10, height 12: the apothem

ANSWER 13

2. Same: lateral area

ANSWER 260

3. Same: total area

ANSWER 360

4. Same: volume

ANSWER 400

5. Cone $r = 5$, $l = 13$: height

ANSWER 12

6. Same cone: curved area

ANSWER 65π

7. Ball of area 144π : radius

ANSWER 6

Medium · the standard question

7 PROBLEMS

1. Cone $r = 5$, $l = 13$: volume

ANSWER 100π

2. Same cone: the sector angle of its net

ANSWER $\approx 138.5^\circ$

3. Frustum $R = 6$, $r = 3$, $h = 4$: volume

ANSWER 84π

4. Same frustum: curved area

ANSWER 45π

5. Ball of area 144π : volume

ANSWER 288π

6. Two similar cones, volumes 27 and 125: the height ratio

ANSWER $3 : 5$

7. Same pair: the surface area ratio

ANSWER $9 : 25$

Hard · stretch and reason

7 PROBLEMS

1. Square pyramid, base 10, height 12: the angle of a lateral edge with the base

ANSWER $\approx 59.5^\circ$

2. Same pyramid: the dihedral angle along a base edge

ANSWER $\approx 67.4^\circ$

3. A cap of height 2 on a ball of radius 6: its volume

ANSWER $\frac{\pi}{3}(4)(18 - 2) = \frac{64\pi}{3} \approx 67.0$

4. A cone and a hemisphere on the same circle have equal volumes. Find $\frac{h}{r}$

ANSWER 2

5. A frustum has radii R and $\frac{R}{2}$ and height h . Find the fraction of the full cone it holds

ANSWER $\frac{7}{8}$

6. A sphere is inscribed in a cone of radius 6 and height 8. Find its radius

ANSWER $r = \frac{48}{16} = 3$

7. A ball is cut by two parallel planes 3 apart on a sphere of radius 10: the zone area

ANSWER 60π

07 Homework

Homework — 4 tasks

Bring the seven-formula page and the concept map to the first lesson of Quarter IV.

1. Rewrite in full every control-work question that lost a mark, naming the slip in the margin.
2. Write the seven formulas of Quarter III on one page in your own hand, each with a small sketch.

3. A cone of radius 9 cm and height 12 cm is cut halfway up by a plane parallel to the base. Find the volume and the curved area of the frustum that remains.
4. Write your target for Quarter IV in one checkable sentence, and date it.